

Inspectable SONIC variables in a MORPHEUS model

The fitted space remains traceable from each named variable to its molecular motion and model role.

HUMAN-READABLE RECORD

CARTESIAN FINGERPRINT

REFINEMENT ROLE

Stre-SALC

$$q_1 = 2^{-1/2}(R_a + R_b)$$

stretch family • length



$-h \quad 0 \quad +h$

SYMMETRY-ALLOWED

candidate active variable

Bend-SALC

$$q_2 = \sum_i c_i \theta_i$$

bend family • angle



$-h \quad 0 \quad +h$

ORBIT-COMPLETE

linked or constrained

RPck

$$q_3 = \sum_i d_i U_i$$

protected ring family • rad



$-h \quad 0 \quad +h$

RANK PROTECTED

retained cage deformation

Stored expansion • family / symmetry / unit • analytic Wilson row • protection and provenance • optional motion trajectory